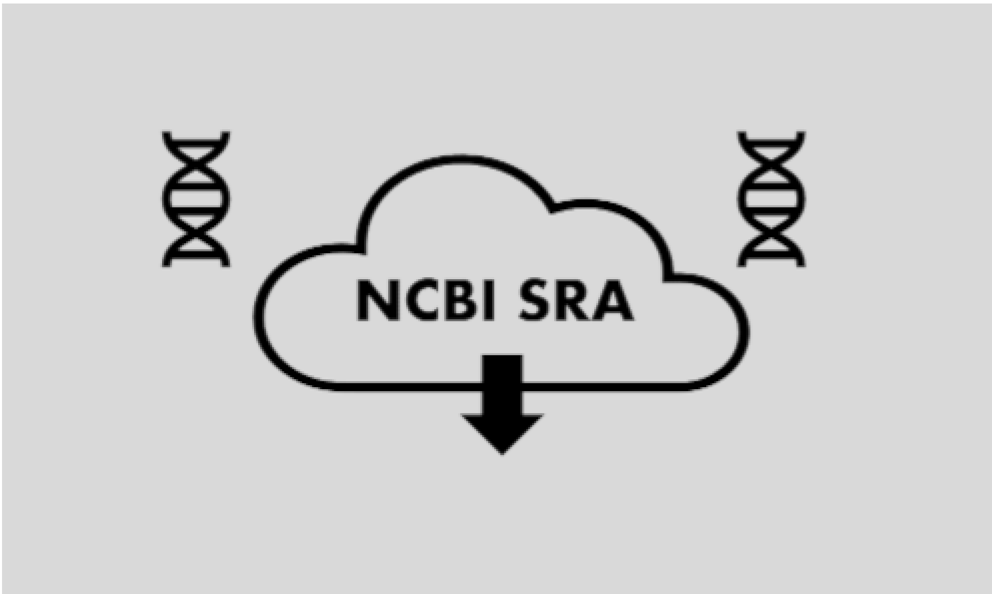
Install Utilities

SRAtoolkit



NOTICE: Shell comands

Google Colab/Jupyter Noetbook using for default phyton as programming language.

It's permite to use another languages, like shell script.

- For this, in Google Colab we use "!" before the code.
- For the Jupyter Notebooks we use a magic cell with `%%bash` before to write our script.

This is a marker to Google Colab understand this is Shell's code. For your personal use, "!" is not necessary.

In [1]:
```
%%bash
# A hashtag is a comment, this parte of the code is not using. However serve to importante ann
# It is a good and common practice in code as a reminder and as a mechanism for reproduction.
# We recommend to always comment your codes and script

echo "Hello, world!"
```

```
Hello, world!
```

Along this Jupyter Notebook you will see diferents command in shell. They will be explained, but here is a small set of the most common commands.

In [2]:
```
%%bash
# show the path of the current working directory

pwd
```

```
/home/jovyan/scNotebooks/Notebooks_EN
```

In [3]:
```
%%bash
# make a folder.
# In programming, folder is called directory. We will use the name directory from here on.
# The name folder could be anything (e.g. folder1, folder_1, etc), try always to name director

mkdir folder
```

In [4]:
```
%%bash
#list of files and directories
# In general, directories has a / in ending (e.g. /Documents/Files/scRNAseq_data/

ls
```

```
folder
Module01_IntroductionNotebooks_Databases.ipynb
Module02_IntroductionToR.ipynb
Module03_Processing_RawSeqs_CellRanger.ipynb
Module04_QualityControl_Clustering_Annotation_FunctionalAnalysis.ipynb
Module05_DatasetIntegration.ipynb
Module06_PseudotimeAnalysis.ipynb
Module07_CellCell_Communication.ipynb
Module08_MultimodalDataAnalysis.ipynb
Module09_TCRsequence.ipynb
Module10_SpatialTranscriptomics.ipynb
Module11_scATACseq.ipynb
Module12_AlternativePoliA.ipynb
Modulo13_PrinciplesFAIR_DataSharing.ipynb
```

In [5]:
```
%%bash
# A command could be follow for arguments. Arguments specify your main code. There are could b
# It's a importante thing to know about a software that you want to use.

ls -l
```

```
total 18848
drwxr-sr-x 2 jovyan users     4096 Feb  9 17:20 folder
-rw-r--r-- 1 jovyan users 14129454 Feb  9 17:11 Module01_IntroductionNotebooks_Databases.ipynb
-rw-r--r-- 1 jovyan users  1620164 Feb  9 17:11 Module02_IntroductionToR.ipynb
-rw-r--r-- 1 jovyan users  1038254 Feb  9 17:11 Module03_Processing_RawSeqs_CellRanger.ipynb
-rw-r--r-- 1 jovyan users  1647000 Feb  9 17:11 Module04_QualityControl_Clustering_Annotation_F
unctionalAnalysis.ipynb
-rw-r--r-- 1 jovyan users   467049 Feb  9 17:11 Module05_DatasetIntegration.ipynb
-rw-r--r-- 1 jovyan users    35056 Feb  9 17:11 Module06_PseudotimeAnalysis.ipynb
-rw-r--r-- 1 jovyan users    36608 Feb  9 17:11 Module07_CellCell_Communication.ipynb
-rw-r--r-- 1 jovyan users    25564 Feb  9 17:11 Module08_MultimodalDataAnalysis.ipynb
-rw-r--r-- 1 jovyan users    34231 Feb  9 17:11 Module09_TCRsequence.ipynb
-rw-r--r-- 1 jovyan users    67197 Feb  9 17:11 Module10_SpatialTranscriptomics.ipynb
-rw-r--r-- 1 jovyan users   106272 Feb  9 17:11 Module11_scATACseq.ipynb
-rw-r--r-- 1 jovyan users    27823 Feb  9 17:11 Module12_AlternativePoliA.ipynb
-rw-r--r-- 1 jovyan users    35541 Feb  9 17:11 Modulo13_PrinciplesFAIR_DataSharing.ipynb
```

In [6]:
```
%%bash
# The command cd is used to move into a directory or out of a directory
# If you want to move to another directory, you can use cd .. to move back forward or only cd

cd
```

In [7]:
```
%%bash
# the command "touch" is used to create a new empty file

touch your_archive
```

In [8]:
```
%%bash
# You also can create a file with text redirecting output of another command with ">" symbol

echo "Hello, World!" > archive
```

In [9]:
```
%%bash
# The command mv is used to move your file to a specific directory
# You only need to specify the path to the directory you want to move your file

mv your_archive folder/
```

In [10]:
```
%%bash
# Also, mv could change names of directories or files

mv folder/ directory/
```

In [11]:
```
%%bash
# The command cat shows the entire contents of your file

cat
```

In [12]:
```
%%bash
# Also, you can concatenate two or more files using this command

cat directory/your_archive archive > archive_3
```

In [13]:
```
%%bash
# The command head shows the first 10 lines of your file
# Like a head of the "cat"

head archive_3
```

```
Hello, World!
```

In [14]:
```
%%bash
# The command tail shows the last 10 lines of your file
# Like a tail of the "cat"

tail
```

In [15]:
```
%%bash
# The command wget is used to download files
# You can put a link next to the command to download data
# e.g. an specific dataset of a database, here is an example of a RNA-seq sample from SRA
```

```
wget -q -O SRR24765940.html https://trace.ncbi.nlm.nih.gov/Traces/sra?run=SRR24765940
```

Installation

To begin, you must download *SRAtoolkit* in order to use the `fastq-dump` tool, which allows you to download SRA files.

The *SRA Toolkit* is a set of tools developed by the National Center for Biotechnology Information (NCBI) to access, download, and manipulate high-throughput data stored in the Sequence Read Archive (SRA)

The `fastq-dump` is one of the tools in the SRA Toolkit. It is used to extract data in FASTQ or FASTA format from SRA accessions. FASTQ is a common format for storing DNA and RNA sequencing data, which includes the sequence reads and their qualities. `fastq-dump` allows users to convert SRA files to FASTQ files, making it easier to process and analyze these data in other bioinformatics tools.

In [16]:
```
%%bash
# The -q is for quit mode, to don't provide all the output of wget, just run silently.
# The --output-document is to specify the name of the file where the downloaded content should
wget -q --output-document sratoolkit.tar.gz https://ftp-trace.ncbi.nlm.nih.gov/sra/sdk/3.1.0/s

# .tar.gz is a file that has been first packaged with tar and then compressed with gzip.
# tar command is used to pack or unpack files on Unix/Linux systems. The name "tar" comes from
# -xzf: Indicates that tar should extract the contents of a gzip-compressed archive.
tar -xzf sratoolkit.tar.gz
```

Then, to test the download, you can see the first three reads from a single SRA accession number.

In [17]:
```
%%bash
# sratoolkit.3.1.0-ubuntu64 is the directorio main, bin is intermediary and fastq-dump is a so
# -X 3: Limits extraction to the first 3 reads of the SRA file. The -X option sets the maximum
# --stdout: Instructs fastq-dump to send output to stdout (standard output) instead of writing
sratoolkit.3.1.0-ubuntu64/bin/fastq-dump -X 3 SRR11537950 --stdout
```

```
Read 3 spots for SRR11537950
Written 3 spots for SRR11537950
```

```
@SRR11537950.1 length=126
TGACTTTCACTAGTACGTCCCGTTAGGCTTAACTATCTTAACAAACTTTGAGTGCAAGAGATTGAAGAGTTCAAATCTGACCAAGATGTTGATGT
TGGATAAGAGAATTCTCTGCTCCCCACCTCT
+SRR11537950.1 length=126
BAAD4@ADCBCCCD@DDDDCDAB@CDDCDACDDCBDCCCCDEDBDDCCCA:BDDDCCDDDECBDECCBBCBEDBBBCACACBC=CB@CCBC4>BB
C<B?BC@CCDEC?D;CCDCCBC@C>BCDC>9
@SRR11537950.2 length=126
GTTCTCGCACCCATTCCGCCGTCAACGGTTTTTTTGAAGTCTTCACAGGCCAGCCAGAATTCAATATTTTCTTCACAGAATTCCGACTTTAAAAA
AGCCCTGAATGCAGCAAGCCCATTTGATGAA
+SRR11537950.2 length=126
BBBCBCDCBDCBDC:ADDCDBABCBCCCBC>ACCBB9>>CCBCB=7CCCB@BCCBABABBBCB2B>@CBBBB?BBCBADB22<C?*DA5AB0
B/?'B=BC::0AB-C-AB@BB%BBA00/<<6@41
@SRR11537950.3 length=126
CCGTGGAGTCTAGTGTCCTAAGAAGTCCCATGCCAATGCAACCTCCCCAGAGAGGGTTTCCCTATCCCCTGCCTGAGGTGCCACAATGACACCCA
TCACTCCTCAGCCCATGGCATTGTCTTTTTT
+SRR11537950.3 length=126
CC>AC?+A?BA<CB?28DC0>,<-A,ADB@;@B2B<>A@?ACC7DBB?@<A=?@BC?<BBBC>A?DBBAAB>?@D?34C@C@<>;??C>D:@DA
<?CB7ADA?BB>6?D?B@%9;A<:1BB>396*
```

Dowload from SRA

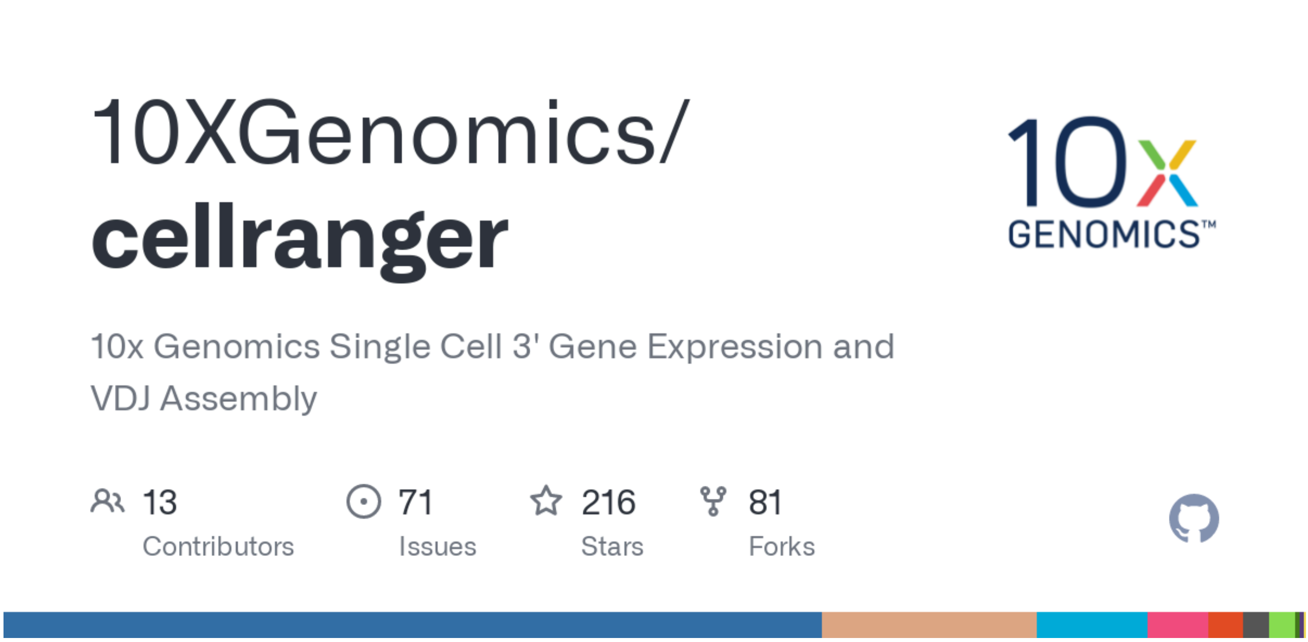
In [18]:
```
%%bash
# We will download SRR11537950 files
# sratoolkit.3.1.0-ubuntu64/bin/fastq-dump: This is the PATH to the fastq-dump executable with
# --split-files: This option instructs fastq-dump to split the paired-end read files into two
# --gzip: This option compresses the output in gzip format, i.e. reduces the file size.
# -X 1000000: Limits the number of reads extracted.
sratoolkit.3.1.0-ubuntu64/bin/fastq-dump --split-files --gzip -X 1000000 SRR11537950

# Create a directory with the same SRA accession number
mkdir -p SRR11537950

# Move everything with "SRR11537950" and ".fastq.gz" information for a folder
mv SRR11537950*.fastq.gz SRR11537950/

# Rename files to apropriate names for Cell Ranger.
mv SRR11537950/SRR11537950_1.fastq.gz SRR11537950/SRR11537950_S1_L001_R1_001.fastq.gz
mv SRR11537950/SRR11537950_2.fastq.gz SRR11537950/SRR11537950_S1_L001_R2_001.fastq.gz
```

```
Read 1000000 spots for SRR11537950
Written 1000000 spots for SRR11537950
```

In [19]:
```
%%bash
# list of archives
ls
```

```
archive
archive_3
directory
Module01_IntroductionNotebooks_Databases.ipynb
Module02_IntroductionToR.ipynb
Module03_Processing_RawSeqs_CellRanger.ipynb
Module04_QualityControl_Clustering_Annotation_FunctionalAnalysis.ipynb
Module05_DatasetIntegration.ipynb
Module06_PseudotimeAnalysis.ipynb
Module07_CellCell_Communication.ipynb
Module08_MultimodalDataAnalysis.ipynb
Module09_TCRsequence.ipynb
Module10_SpatialTranscriptomics.ipynb
Module11_scATACseq.ipynb
Module12_AlternativePoliA.ipynb
Modulo13_PrinciplesFAIR_DataSharing.ipynb
sratoolkit.3.1.0-ubuntu64
sratoolkit.tar.gz
SRR11537950
SRR24765940.html
```

Cell Ranger



What is Cell Ranger?

Cell Ranger is a software suite developed by 10x Genomics for analyzing single-cell RNA sequencing (scRNA-seq) data generated from their Chromium platform. It processes the raw sequencing data into meaningful insights, including gene expression matrices, cell clustering, and other downstream analyses.

Note:

> Cell Ranger requires more than 12GB of RAM to run, it does not work with Colab.

Download

Download Cellranger

- Update the link if it has expired and then the relevant genomic reference.

In [ ]:
```
%%bash
# -O: name of output

wget -q -O cellranger-10.0.0.tar.gz \
"https://cf.10xgenomics.com/releases/cell-exp/cellranger-10.0.0.tar.gz?Expires=1770474025&Key-
```

In [ ]:
```
%%bash
tar -xzf cellranger-10.0.0.tar.gz
```

In [ ]:
```
%%bash
wget "https://cf.10xgenomics.com/supp/cell-exp/refdata-gex-GRCh38-2024-A.tar.gz"
```

In [ ]:
```
%%bash
tar -xzf refdata-gex-GRCh38-2024-A.tar.gz
```

Alternatively, you can also download version 10.0.0 from a GitHub repository we created as a backup, but first you need to remove the "#" symbol to uncomment it, as it's not necessary to run it if you click the download link from the 10x website.

In [ ]:
```
%%bash
#wget https://github.com/integrativebioinformatics/scNotebooks/blob/main/scNotebooks-Resources
```

IMPORTANT:

If your current version of cellranger is different from the one used in this Jupyter Notebook, you will need to update it in the code directory or download the backup from GitHub.

> Example: cellranger-9.0.1 --> cellranger-10.0.0

Installation

To work with certain organisms not available in the resources provided by 10x, such as bacteria, plants, and viruses, it is necessary to create a reference for CellRanger:

- cellranger-10.0.0/bin/cellranger is the path to the software
- The mkref command is used to create custom reference packages
- --genome: Creates a directory for the output files
- --fasta: Specifies the reference genome
- /genome/genome.fa: Is the path to the genome in .fa format

- --genes: Specifies the reference GTF file
- /gtf/genome.annotation.gtf: Path to the annotation file

Here's a small example of how to generate a reference, in this case for chromosome 22, from the 10x reference human genome.

First, we install samtools to manipulate the FASTA genome file and select chromosome 22.

In [ ]:
```
%%bash
#apt-get update
apt-get install samtools -y #Install samtools
```

The `samtools faidx` function in particular extracts chromosome 22, and then we redirect the output to a file called GRCh38.CHR22.fa.

In [ ]:
```
%%bash
samtools faidx refdata-gex-GRCh38-2024-A/fasta/genome.fa chr22 > GRCh38.CHR22.fa
```

In [ ]:
```
%%bash
gunzip -c refdata-gex-GRCh38-2024-A/genes/genes.gtf.gz | grep "chr22" | gzip > CHR22.genes.gtf

# gunzip is for descompress archives gzip
# -c: indicates that the decompressed output should be sent to standard output (stdout) instea
# refdata-gex-GRCh38-2024-A/genes/genes.gtf: is the path to the compressed GTF file that conta
# grep "chr22": filters the lines that contain "chr22", extracting them
# gzip > CHR22.genes.gtf.gz: compresses the filtered lines and saves them to a new file called
# | (pipe) is an operator that takes the output of one command and uses it as input to another
```

Finally, we have the necessary files to generate the new reference and we can run cellranger.

In [ ]:
```
%%bash
cellranger-10.0.0/bin/cellranger mkref --genome=GRCh38-CHR22 --fasta=GRCh38.CHR22.fa --genes=C

# cellranger-10.0.0/bin/cellranger: is the path to the Cell Ranger executable.
# --genome=GRCh38-CHR22: Name of the reference genome being created.
# --fasta=GRCh38.CHR22.fa: Path to the FASTA file containing the reference genome sequences.
# --genes=CHR22.genes.gtf.gz: Path to the GTF file containing the genomic annotations.
```

Now, we have our new reference available, which we can work with.

DO NOT RUN!

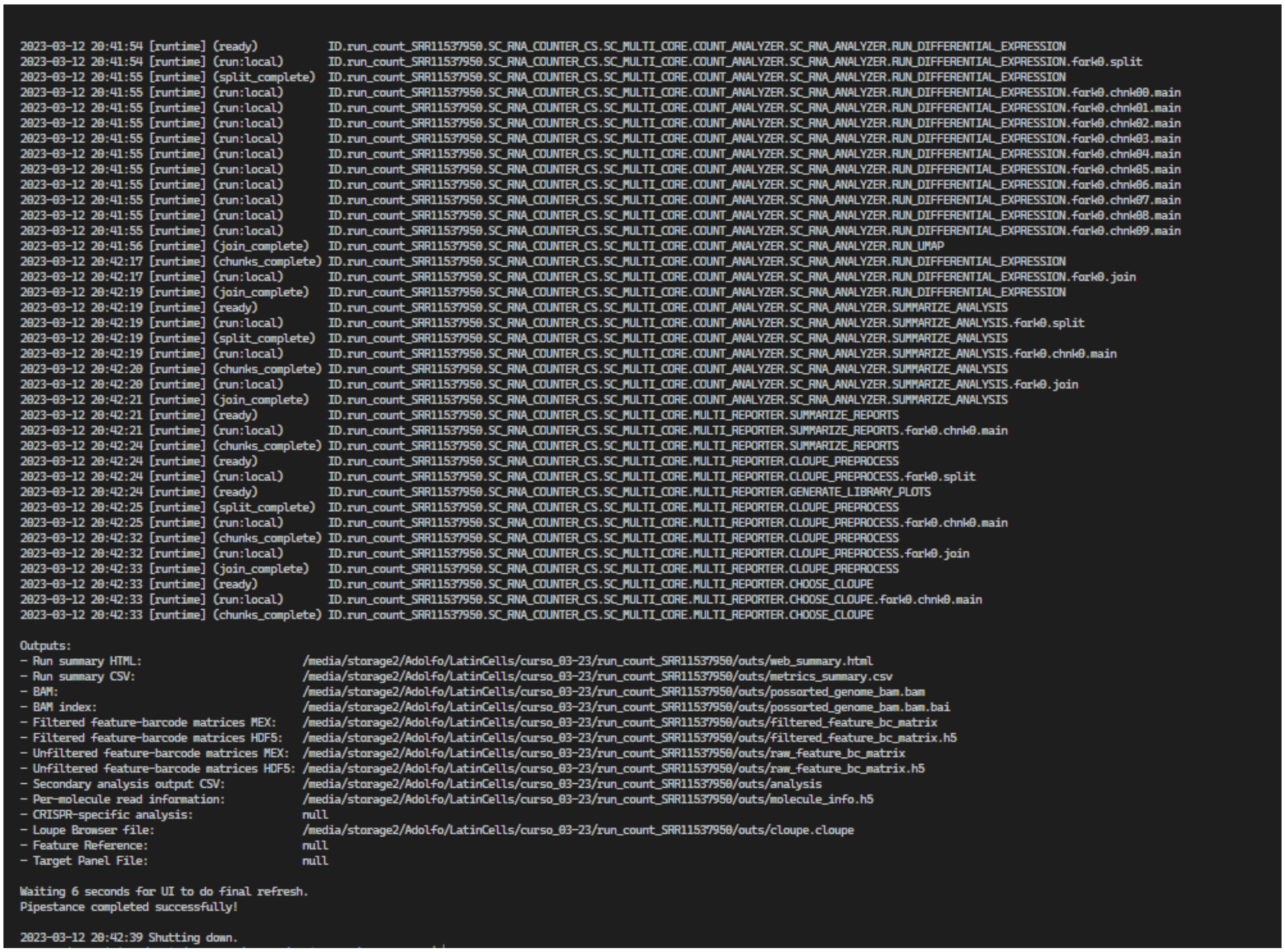
`cellranger-10.0.0/bin/cellranger` is a path to the software

- `count` is used to count messenger RNA (mRNA) molecules
- `--id` is a directory name for outputs
- `--fastqs` path to files to analyse
- `--sample` files to analyse

- `--transcriptome` is the reference

In [ ]:
```
%%bash
#DO NOT RUN!
#This \ break a line and continue the code in another line
cellranger-10.0.0/bin/cellranger count --id=run_count_SRR11537950 \
    --fastqs=SRR11537950 --sample=SRR11537950 \
    --transcriptome=refdata-gex-GRCh38-2020-A
#--localcores=2 --localmem=20
```

Cellranger outputs running one sample



```
2023-03-12 20:41:54 [runtime] (ready)           ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION
2023-03-12 20:41:54 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.split
2023-03-12 20:41:55 [runtime] (split_complete)  ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk00.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk01.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk02.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk03.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk04.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk05.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk06.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk07.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk08.main
2023-03-12 20:41:55 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.chnk09.main
2023-03-12 20:41:56 [runtime] (join_complete)   ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_UMAP
2023-03-12 20:42:17 [runtime] (chunks_complete) ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION
2023-03-12 20:42:17 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION.fork0.join
2023-03-12 20:42:19 [runtime] (join_complete)   ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.RUN_DIFFERENTIAL_EXPRESSION
2023-03-12 20:42:19 [runtime] (ready)           ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS
2023-03-12 20:42:19 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS.fork0.split
2023-03-12 20:42:19 [runtime] (split_complete)  ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS
2023-03-12 20:42:19 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS.fork0.chnk0.main
2023-03-12 20:42:20 [runtime] (chunks_complete) ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS
2023-03-12 20:42:20 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS.fork0.join
2023-03-12 20:42:21 [runtime] (join_complete)   ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.COUNT_ANALYZER.SC_RNA_ANALYZER.SUMMARIZE_ANALYSIS
2023-03-12 20:42:21 [runtime] (ready)           ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.SUMMARIZE_REPORTS
2023-03-12 20:42:21 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.SUMMARIZE_REPORTS.fork0.chnk0.main
2023-03-12 20:42:24 [runtime] (chunks_complete) ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.SUMMARIZE_REPORTS
2023-03-12 20:42:24 [runtime] (ready)           ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS
2023-03-12 20:42:24 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS.fork0.split
2023-03-12 20:42:24 [runtime] (ready)           ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.GENERATE_LIBRARY_PLOTS
2023-03-12 20:42:25 [runtime] (split_complete)  ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS
2023-03-12 20:42:25 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS.fork0.chnk0.main
2023-03-12 20:42:32 [runtime] (chunks_complete) ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS
2023-03-12 20:42:32 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS.fork0.join
2023-03-12 20:42:33 [runtime] (join_complete)   ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CLOUPE_PREPROCESS
2023-03-12 20:42:33 [runtime] (ready)           ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CHOOSE_CLOUPE
2023-03-12 20:42:33 [runtime] (run:local)       ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CHOOSE_CLOUPE.fork0.chnk0.main
2023-03-12 20:42:33 [runtime] (chunks_complete) ID.run_count_SRR11537950.SC_RNA_COUNTER_CS.SC_MULTI_CORE.MULTI_REPORTER.CHOOSE_CLOUPE

Outputs:
- Run summary HTML:                         /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/web_summary.html
- Run summary CSV:                          /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/metrics_summary.csv
- BAM:                                      /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/possorted_genome_bam.bam
- BAM index:                                /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/possorted_genome_bam.bam.bai
- Filtered feature-barcode matrices MEX:    /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/filtered_feature_bc_matrix
- Filtered feature-barcode matrices HDF5:   /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/filtered_feature_bc_matrix.h5
- Unfiltered feature-barcode matrices MEX:  /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/raw_feature_bc_matrix
- Unfiltered feature-barcode matrices HDF5: /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/raw_feature_bc_matrix.h5
- Secondary analysis output CSV:            /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/analysis
- Per-molecule read information:            /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/molecule_info.h5
- CRISPR-specific analysis:                 null
- Loupe Browser file:                       /media/storage2/Adolfo/LatinCells/curso_03-23/run_count_SRR11537950/outs/cloupe.cloupe
- Feature Reference:                        null
- Target Panel File:                        null

Waiting 6 seconds for UI to do final refresh.
Pipestance completed successfully!

2023-03-12 20:42:39 Shutting down.
```

> The Cell Ranger software strives to maintain compatibility with common analysis tools by using standard output file formats whenever possible.

> For example, the barcoded BAM files can be viewed in standard genome browsers such as IGV to verify alignment quality and other features.

The cellranger count pipeline outputs an interactive summary HTML file named web_summary.html that contains summary metrics and automated secondary analysis results. Let's check it out!

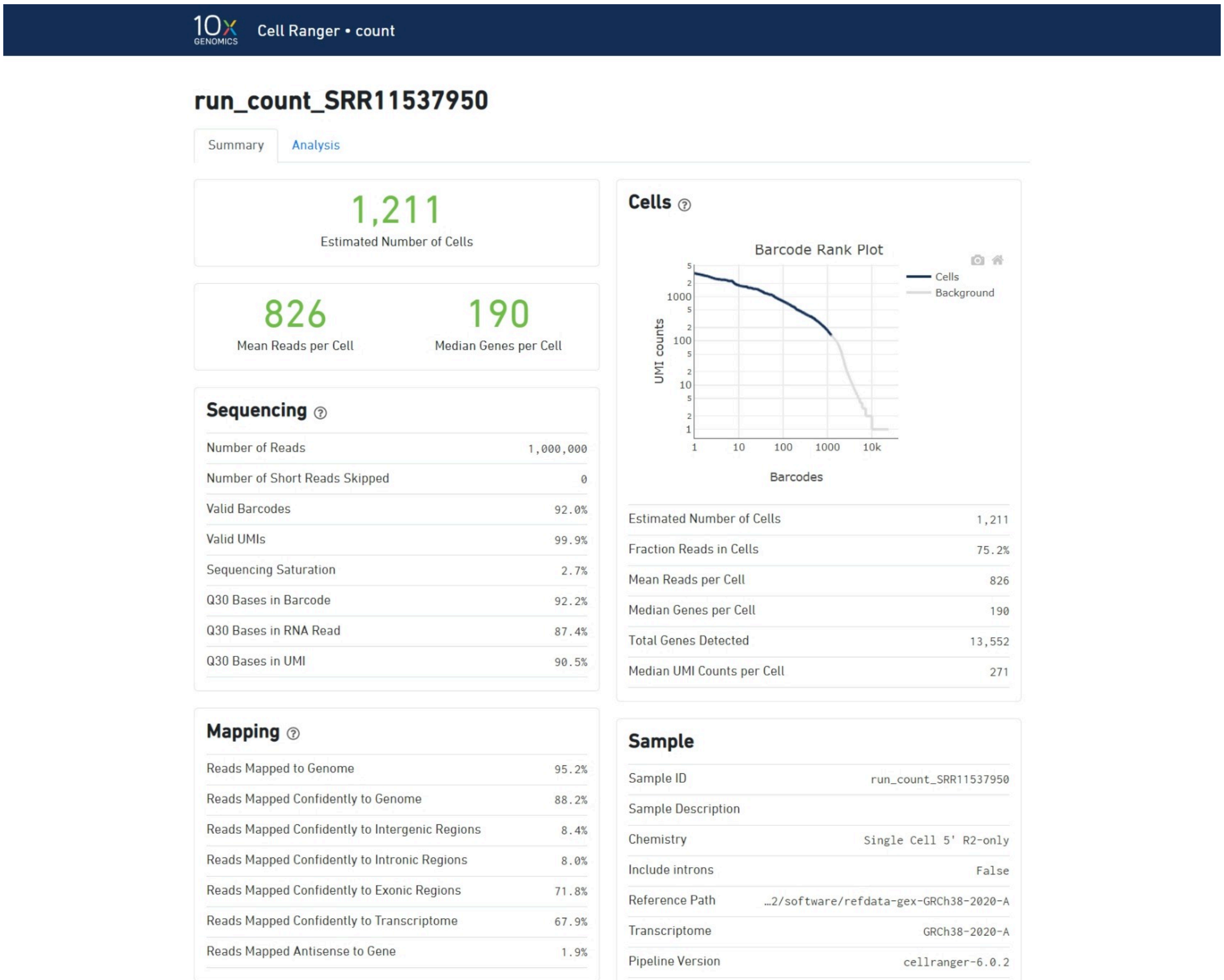
https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/output/summary

Understand the summary output:

- **Green** text indicates that the key metrics are in the expected range.
- **Red/Yellow** text indicates errors/warnings.

Barcode Rank Plot:

A steep drop-off is indicative of good separation between the cell-associated barcodes and the barcodes associated with empty GEMs.


10x GENOMICS Cell Ranger • count

run_count_SRR11537950

Summary Analysis

1,211
Estimated Number of Cells

826
Mean Reads per Cell

190
Median Genes per Cell

Sequencing

Number of Reads	1,000,000
Number of Short Reads Skipped	0
Valid Barcodes	92.0%
Valid UMIs	99.9%
Sequencing Saturation	2.7%
Q30 Bases in Barcode	92.2%
Q30 Bases in RNA Read	87.4%
Q30 Bases in UMI	90.5%

Mapping

Reads Mapped to Genome	95.2%
Reads Mapped Confidently to Genome	88.2%
Reads Mapped Confidently to Intergenic Regions	8.4%
Reads Mapped Confidently to Intronic Regions	8.0%
Reads Mapped Confidently to Exonic Regions	71.8%
Reads Mapped Confidently to Transcriptome	67.9%
Reads Mapped Antisense to Gene	1.9%

Cells

Estimated Number of Cells	1,211
Fraction Reads in Cells	75.2%
Mean Reads per Cell	826
Median Genes per Cell	190
Total Genes Detected	13,552
Median UMI Counts per Cell	271

Sample

Sample ID	run_count_SRR11537950
Sample Description	
Chemistry	Single Cell 5' R2-only
Include introns	False
Reference Path	…2/software/refdata-gex-GRCh38-2020-A
Transcriptome	GRCh38-2020-A
Pipeline Version	cellranger-6.0.2

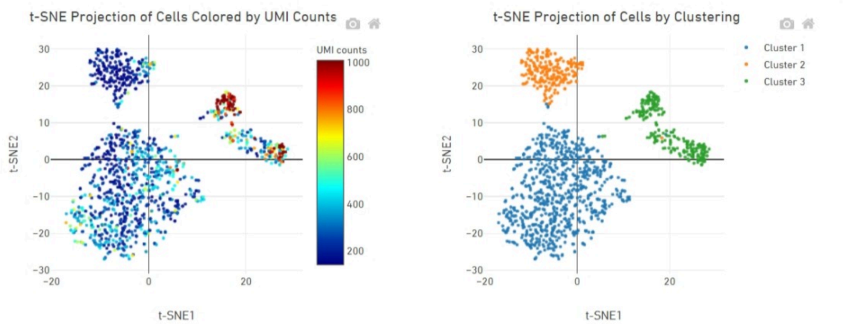
10x GENOMICS Cell Ranger • count

run_count_SRR11537950

Summary Analysis

t-SNE Projection

Clustering Type: Graph-based

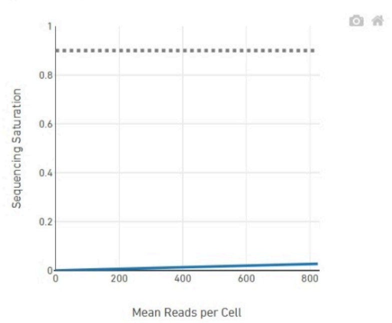


Top Features by Cluster (Log2 fold-change, p-value)

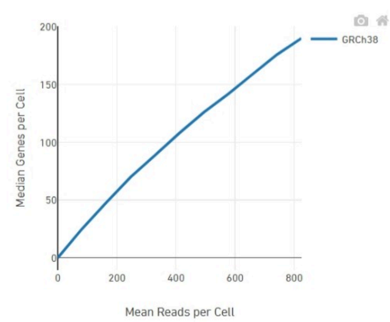
Feature		Cluster 1		Cluster 2		Cluster 3	
ID	Name	L2FC	p-value	L2FC	p-value	L2FC	p-value
ENSG00000108691	CCL2	7.54	2e-69	-5.72	2e-9	-7.64	1e-55
ENSG00000158869	FCER1G	6.84	2e-51	-4.39	6e-6	-7.43	2e-42
ENSG00000135047	CTSL	6.74	5e-61	-5.42	7e-9	-6.58	9e-48
ENSG00000108700	CCL8	6.68	1e-56	-8.17	2e-11	-6.09	2e-41
ENSG00000216490	IFI30	5.22	1e-48	-4.14	1e-6	-5.05	8e-38
ENSG00000087086	FTL	4.57	2e-43	-3.69	6e-6	-4.36	2e-33
ENSG00000122862	SRGN	3.95	9e-30	-0.91	7e-1	-9.74	2e-50
ENSG00000164733	CTSB	3.59	2e-24	-4.12	2e-5	-3.12	3e-16
ENSG00000117984	CTSD	3.55	2e-26	-4.87	2e-7	-3.03	4e-17
ENSG00000101439	CST3	2.26	3e-12	-4.20	5e-6	-1.72	1e-6

Previous Page 1 of 8 10 rows Next

Sequencing Saturation



Median Genes per Cell



The significate of t-SNE, clusters, features and others informations will be explain in others Modules.